

Birla Institute of Technology & Science, Pilani, Rajasthan  
First Semester 2025-2026  
Tutorial-7

Course: ECE/EEE F311 Communication Systems  
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

Date: 25-09-2025

## Objectives

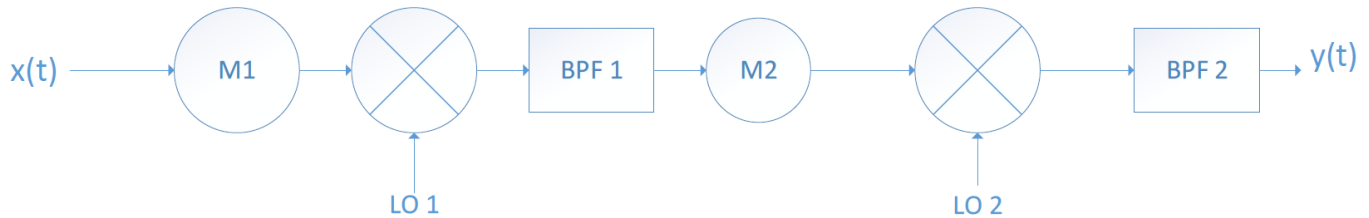
- FM Generation, Superhetrodyne Receiver

## Tutorial Problems

1. An FM signal  $x(t) = 20 \cos(2\pi 10^5 t + 0.3 \sin(2\pi 100 t))$  is passed through multipliers  $M_1$  and  $M_2$ . Local oscillators with frequencies  $f_{LO1}$  and  $f_{LO2}$ , and bandpass filters BPF 1 and BPF 2 generate another FM signal

$$y(t) = 20 \cos(2\pi 10^7 t + 15 \sin(2\pi 100 t)).$$

Find the specifications of  $M_1$ ,  $M_2$ ,  $f_{LO1}$ ,  $f_{LO2}$ , and the bandwidth and carrier frequencies for BPF 1 and BPF 2. The design should satisfy the constraints: multipliers  $M_1 > 1$ ,  $M_2 > 1$ , and local oscillators  $f_{LO1}$ ,  $f_{LO2}$  each having a maximum frequency of 10 MHz.



2. A transmitter transmits an AM signal with a carrier frequency of 1500 kHz. When an inexpensive radio receiver (which has a poor selectivity in its RF-stage bandpass filter) is tuned to 1500 kHz, the signal is heard loud and clear. This same signal is also heard (not as well) at another dial setting. State, with reasons, at what frequency you will hear this station. The IF frequency is 455 kHz.

## Solution

1. Multiple solutions depending on choice of  $M_1$  and  $M_2$ . We can take  $M_1 = 25$  and  $M_2 = 2$ , which gives  $\beta = 15$ .

First stage:

The carrier frequency after  $M_1 = 100\text{KHz} \times 25 = 2.5\text{MHz}$  and  $\beta = 0.3 \times 25 = 7.5$ . We take  $L_{O1}$  at 1.5MHz. Input to BPF 1: FM signal at carrier 4MHz, bandwidth of the message signal 100 Hz. Using Carson's rule  $2(\beta + 1)f_m = 2(8.5) \times 100 = 1.7\text{KHz}$ .

BPF 1: BW = 1.7 KHz, center frequency 4 MHz.

Second stage:

FM signal with carrier frequency (after  $M_2$ ): 8 MHz with  $\beta = 2 \times 7.5 = 15$  We take  $L_{O2}$  at 2 MHz to get final FM signal at 10 MHz.

BPF 2: BW =  $2(15 + 1) \times 100 = 3.2$  KHz, center frequency 10 MHz.

2. Image frequency:  $f_c + 2f_{IF} = 1500 + 2 \times 455 = 2410$  KHz.